Review: Compound Interest and e

The formula for continuously compounded interest is an example of *exponential growth*.

Definition The number e is defined as follows:

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828...$$

Interest Rates If we put $\$A$ in the bank at annual interest rate r (expressed as a decimal), continuously compounded, then after t years, we have $\$Ae^{rt}$.

Preliminaries: Bacterial Growth

A bacterial culture was placed in an environment that allows it unrestricted growth. The number of bacteria was measured every hour for 8 hours, resulting in the following data:

Time (hours)	0	1	2	3	4	5	6	7	8
# Bacteria	15	49	161	525	1719	5625	18406	60226	197059
Δ Bacteria	X	$49 - 15 = 34$							
Ratio # Bac		$49/15 \approx 3.27$							

- You can find this data in a spreadsheet here. Make a copy of it.
- In column B, compute the successive differences between the number of bacteria.
- In column C, compute the successive ratios between the number of bacteria (same as difference, but divide instead of subtract this time).
- Copy your results into the table above.

Question Explain why your results shows that bacterial growth is not linear.

At $t = 0$, there are 15 bacteria. After one hour, at $t = 1$ hour, the number of bacteria is

$$49 \approx 15 \times _____.$$

After two hours, at $t = 2$ hours, the number of bacteria is

$$161 \approx 49 \times _____ = 15 \times _____ \times _____ = 15 \times (_____)^2.$$

Every hour, the number of bacteria increases by a ratio of approximately 3.27. Therefore, after t hours, the number of bacteria is given by

$$B(t) = 15(3.27)^t.$$

Suppose we want to know how long it took for the number of bacteria to double from 15 to 30 bacteria. Looking back at the table, we can see that by time $t = 1$ hour, we already had more than that. So the time we're looking for must be between $t = 0$ hours and $t = 1$ hour.

Question Write down an equation who solution would tell you the time at which there were 30 bacteria. Then take one step toward solving it.

At this point, we do not have the tools to go further (we'll get back to that!). For now, try plugging in $t = 0.5$ hours as a first guess, then decide if you need to go a bit higher or a bit lower. Keep doing this until you get a solution to three decimal places. Fill in your guesses in the table below.

Time (hours)	0.5						
3.27^t							

The bacteria double from 15 to 30 at time $t =$ _____ hours.

Question At $t = 1$ hours, there are 49 bacteria. Write down an equation whose solution will give you the time at which there are double that, 98 bacteria. We know that the solution must be between one and two hours. Once again, take one step toward solving the equation, then do a similar guess-and-check to get the value of t. How long does it take the bacteria to double from 49 to 98 bacteria?

Time (hours)	1.5						
3.27^t							

It takes _____ hours.

If you did this right, you should have found that it takes the same amount of time to double from 49 to 98 bacteria is it took to double from 15 to 30 bacteria. In fact, for all exponential functions, *doubling time* is constant.

Definition A (growing) ***exponential function*** is a function of the form

$$f(x) = P_0 a^x,$$

where P_0 is the *initial value* and $a > 1$ is the *growth factor*.

Definition The *doubling time* of an exponentially increasing function is the time required for the function to double in value.

1. What is the doubling time of the value of a bank account with interest rate 5%, □Regrade compounded continuously? (Note: if the interest is 5%, then $r = 0.05$. Guess and check again. Start your guesses with $t = 10$ years and go from there.)

Time (years)	10	15					
$e^{0.05t}$							

Converting from e^{kx} to a^x

2. Recall that $a^{bc} = (a^b)^c = (a^c)^b$. Use this to convert $Ae^{0.05t}$ to the form Aa^t. Write a □Regrade to four decimal places.

Forms of Exponential Functions Any exponential function can be written as

$$f(x) = P_0 a^x \text{ or } f(x) = P_0 e^{kx}.$$

The forms are equivalent. We will learn to convert back and forth between them.

P_0 is called the *initial value*. k is the *continuous growth rate*.

Exponential Decay

Not all exponential functions grow! If $0 < a < 1$ (or $k < 0$ in the e^{kx} form), the function decreases as x increases. We say that it *decays* instead of grows.

Radioactive Decay

Radioactive elements decay with a constant *half-life*. In other words, if we start with n atoms of a radioactive substance with half-life T, we expect to be left with _____ atoms of the substance after time T has passed.

Example

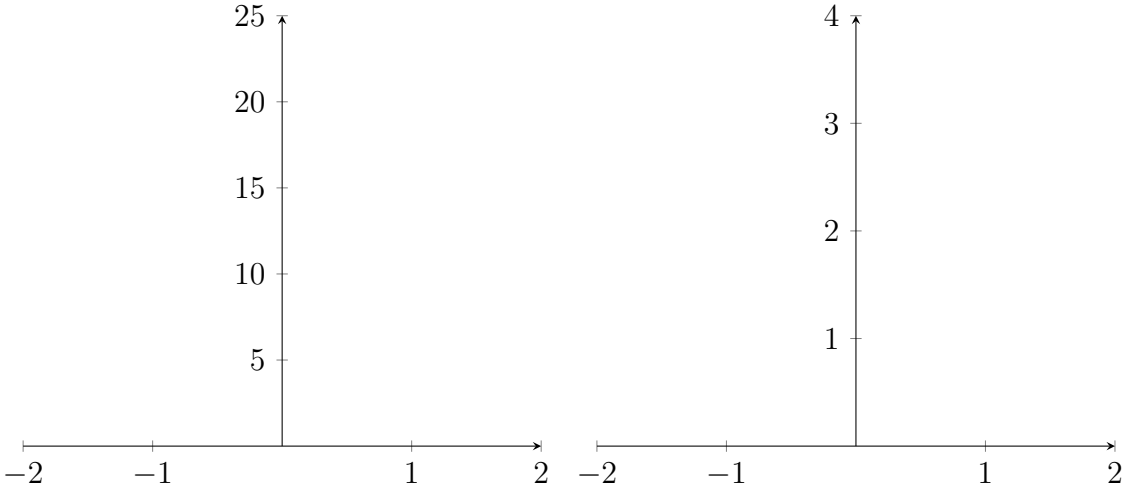
3. □Regrade Proctactinium-234, a product of the radioactive decay of uranium-238, has a half-life of 1.2 minutes. Suppose we have 10 moles of proctactinium-234 at time $t = 0$.

(a) How much do we have after 1.2 minutes? 2.4 minutes?

(b) Write down a formula of the form $P = P_0 a^t$ for the number of moles left after t minutes. (Hint: you already know three values of P. You really only need two...)

Graphing Exponential Functions

4. □Regrade (a) Using Geogebra (or otherwise), graph the functions $f(x) = a^x$ for $a = 1.5$, 2, 3, and 5. Copy the graphs on the first set of axes and label them. Do the same for 0.95, 0.9, 0.8, and 0.5 on the right axes.



(b) Fill in the blanks:

- All graphs of exponential functions are concave _____.
- If $a > 1$, the graphs _____. If $a < 1$, the graphs _____.
- If $a > 1$, the larger a the _____ the graph grows.
- If $a < 1$, the closer a is to 0, the _____ the graph decays.